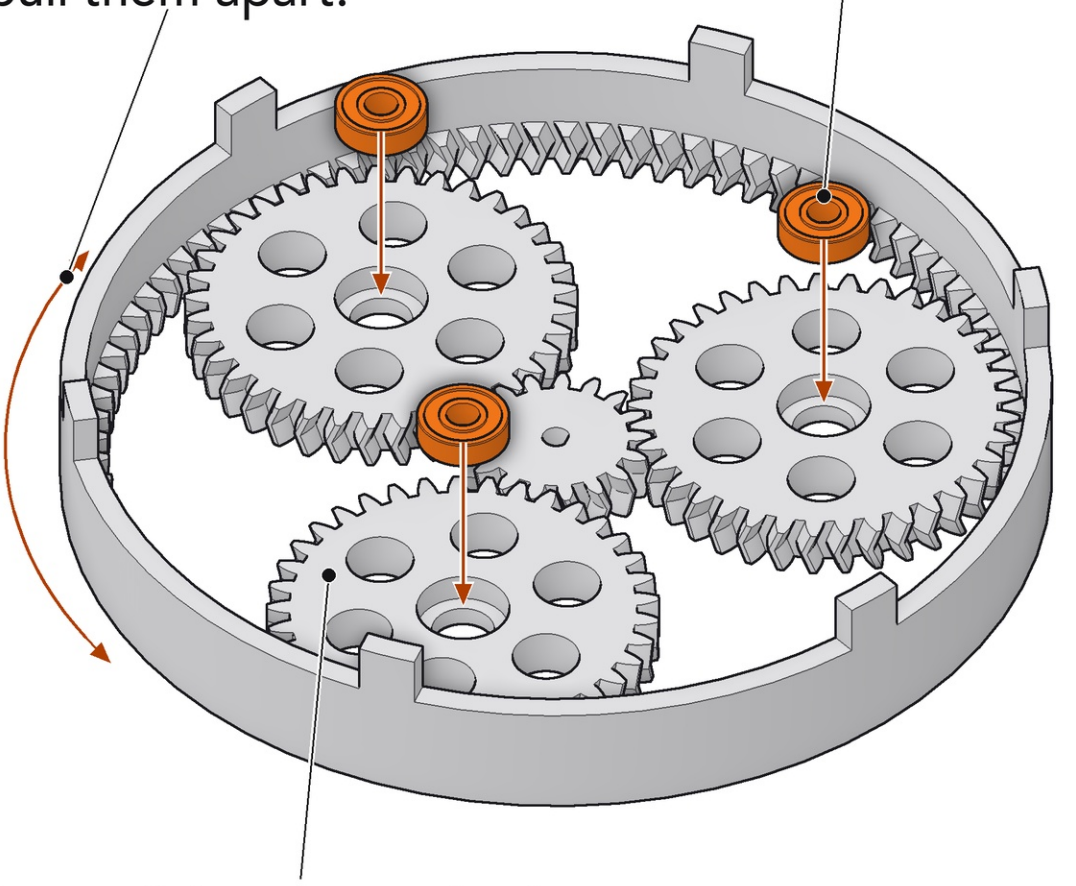
Step 1

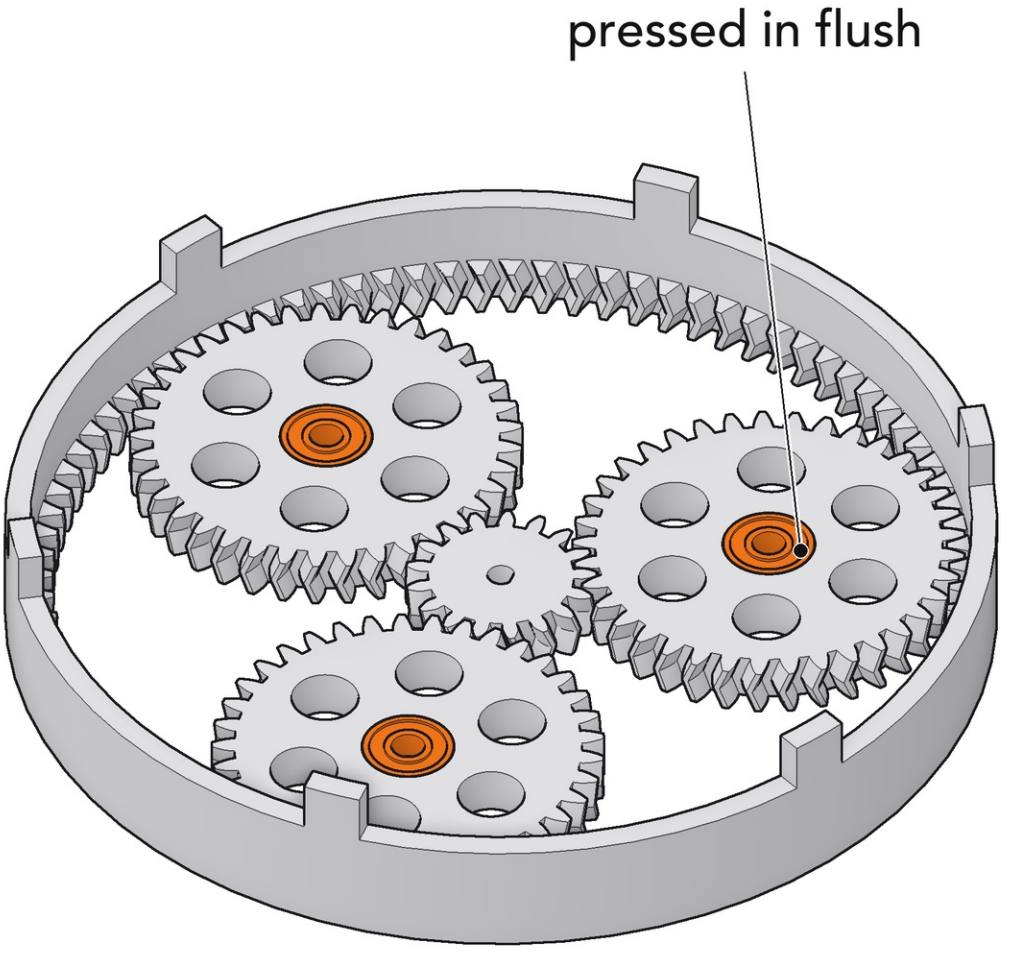
Free the gears, then a 608 bearing into each planet.

1. FIRST twist the ring against the sun until all 5 gears turn. Do not pull them apart.

2. 608 bearing, x3: press one into the front of each planet



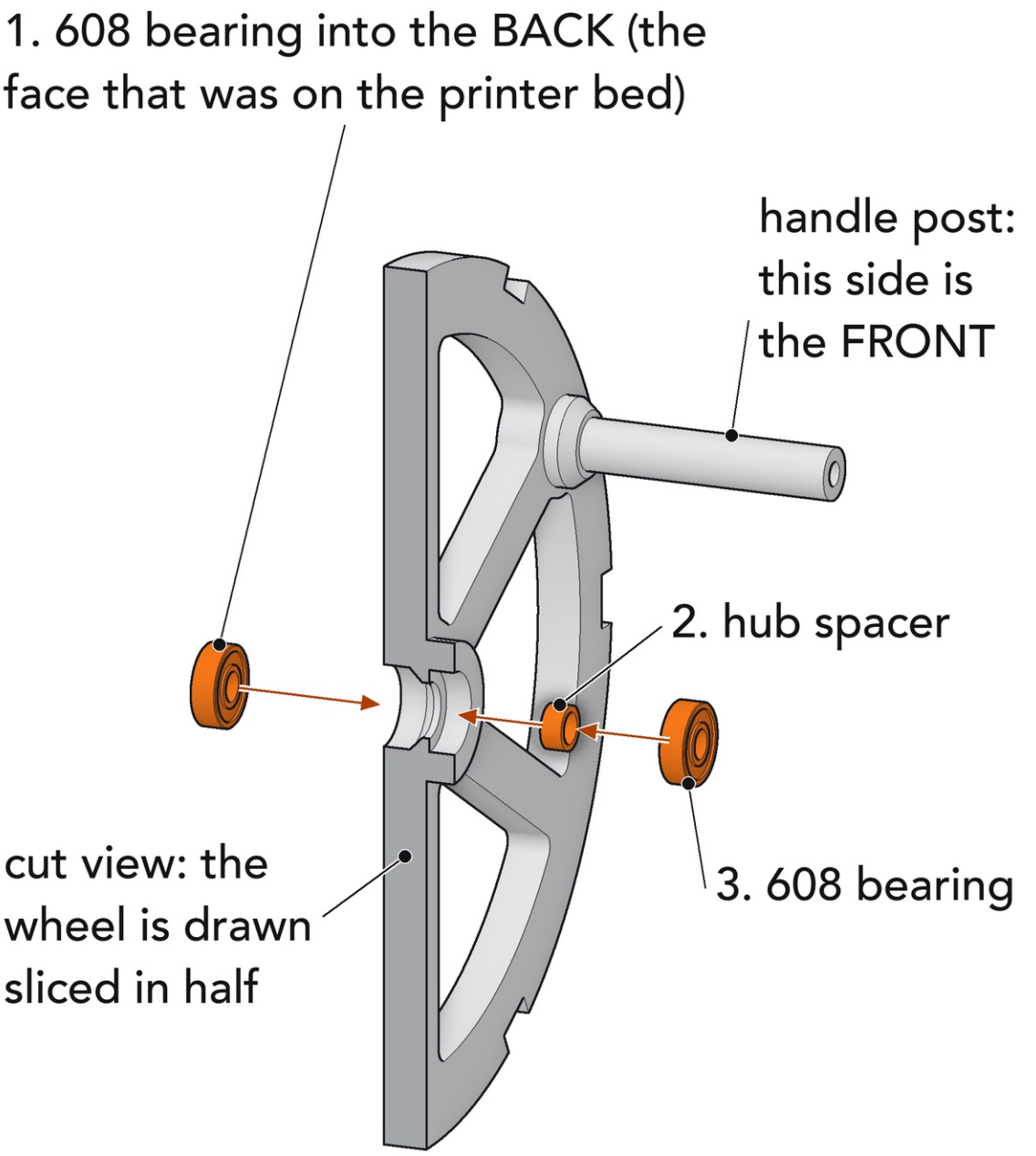
front of the planet: the face that was up on the printer



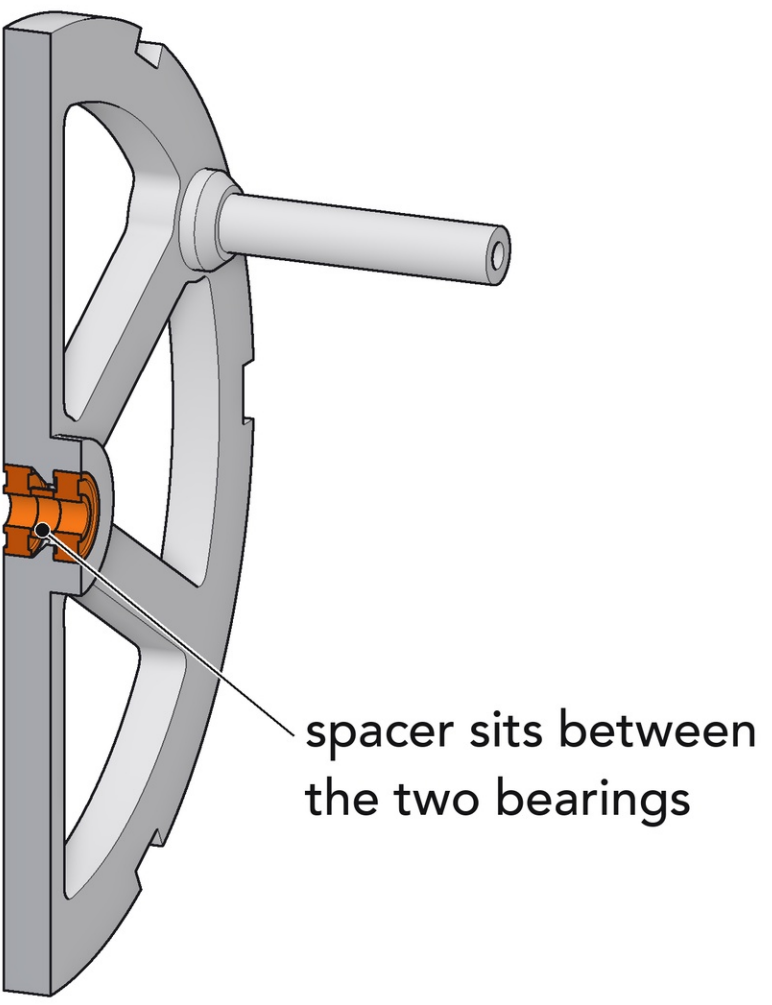
all five gears still turn

Step 2

Wheel hub: bearing in the back, then spacer, then bearing.



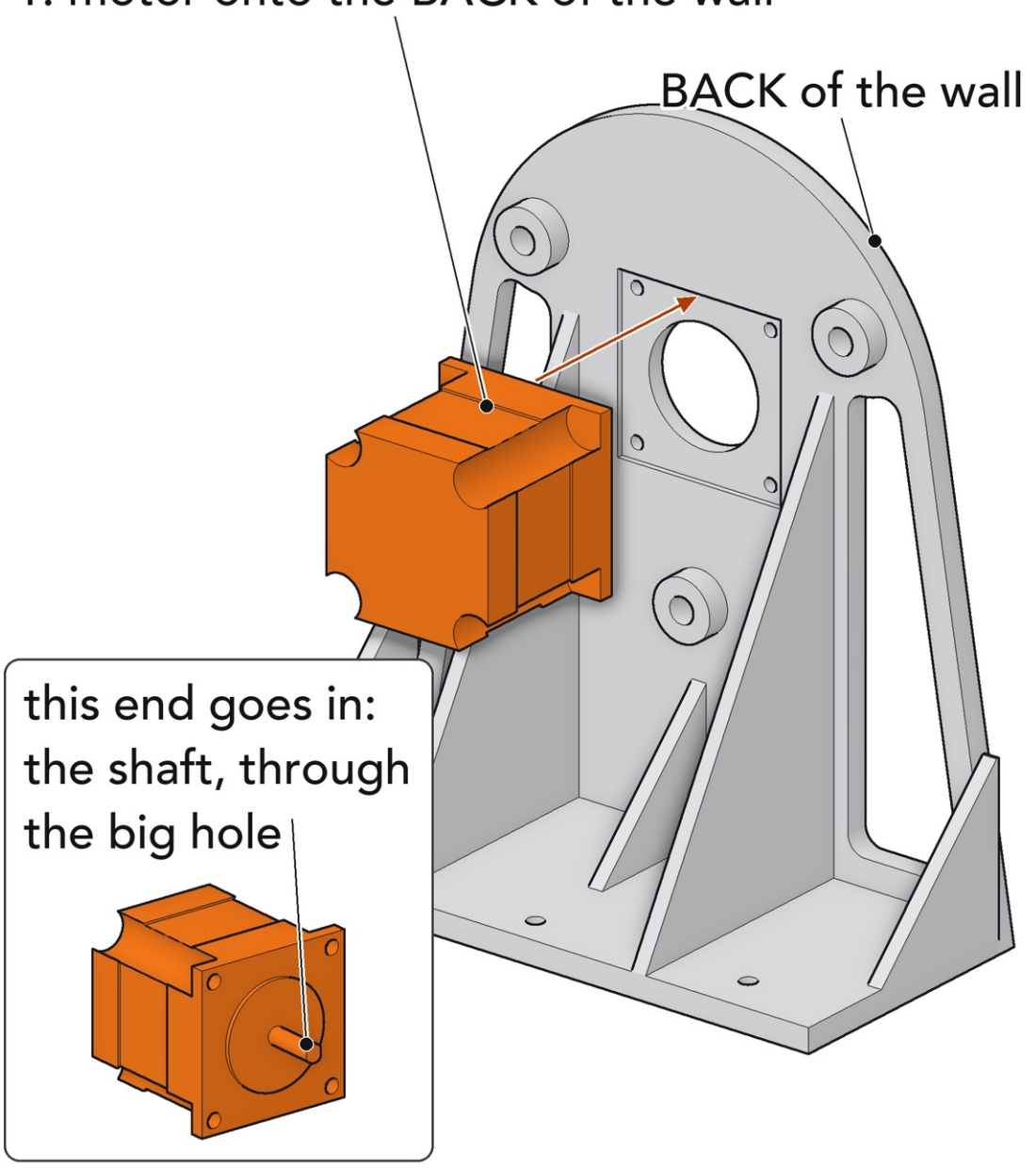
to press bearing 1 in: lay the wheel front down, the post hanging over the table edge



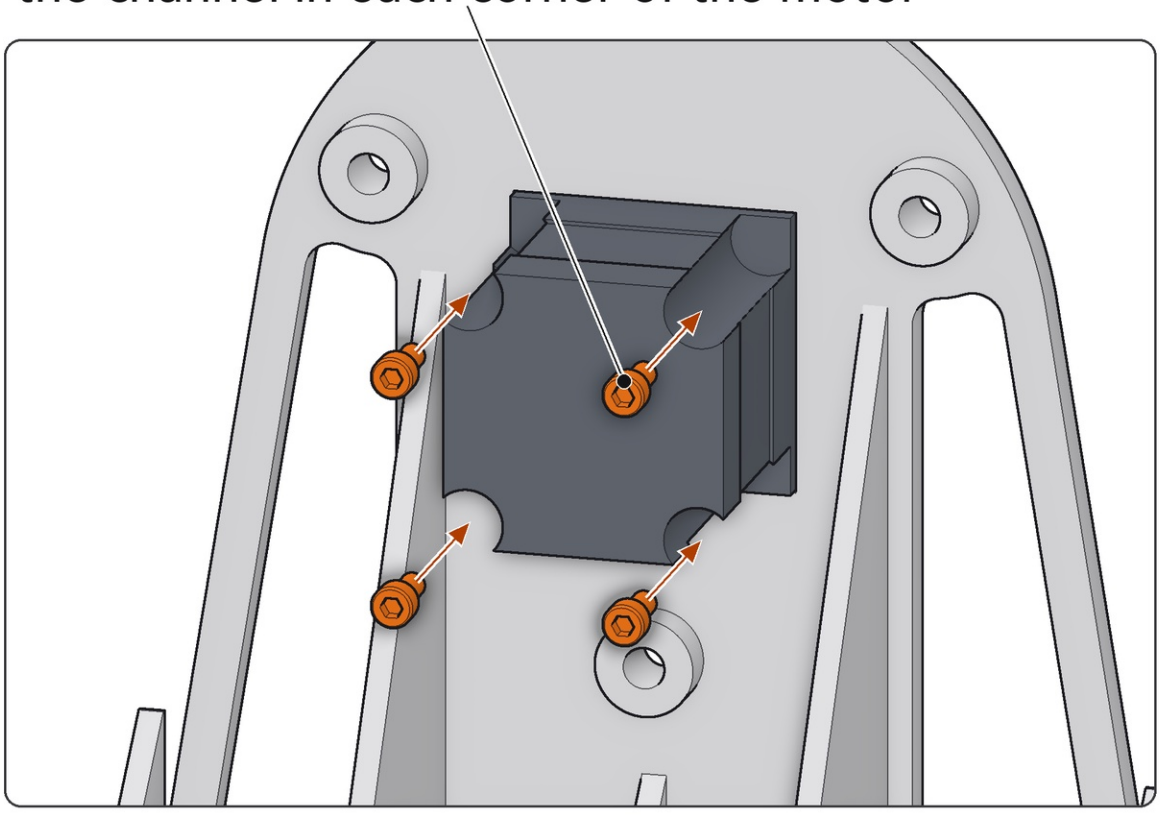
Step 3

Bolt the motor onto the back of the wall.

1. motor onto the BACK of the wall



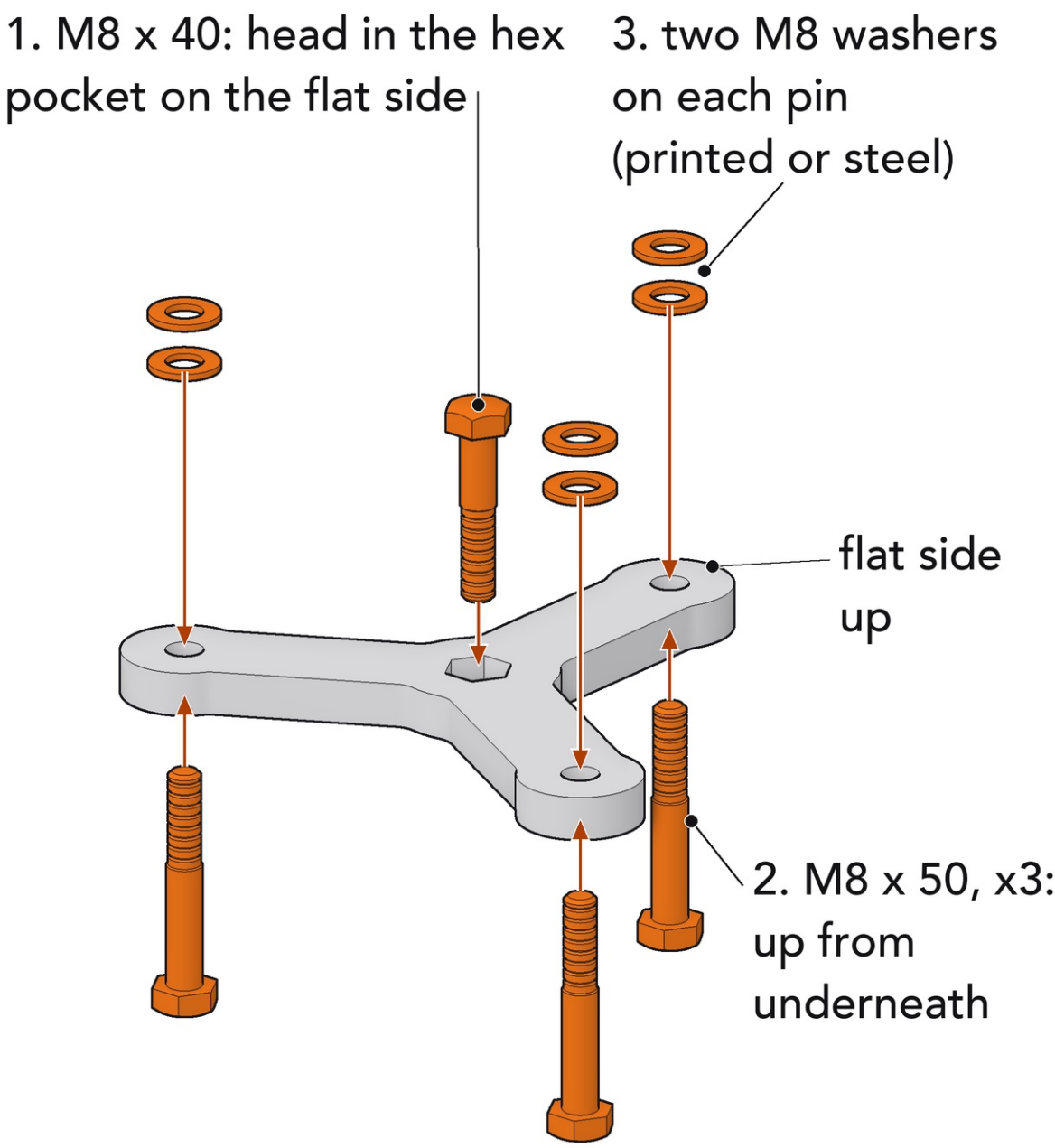
2. M5 x 12 socket cap, x4: one down the channel in each corner of the motor

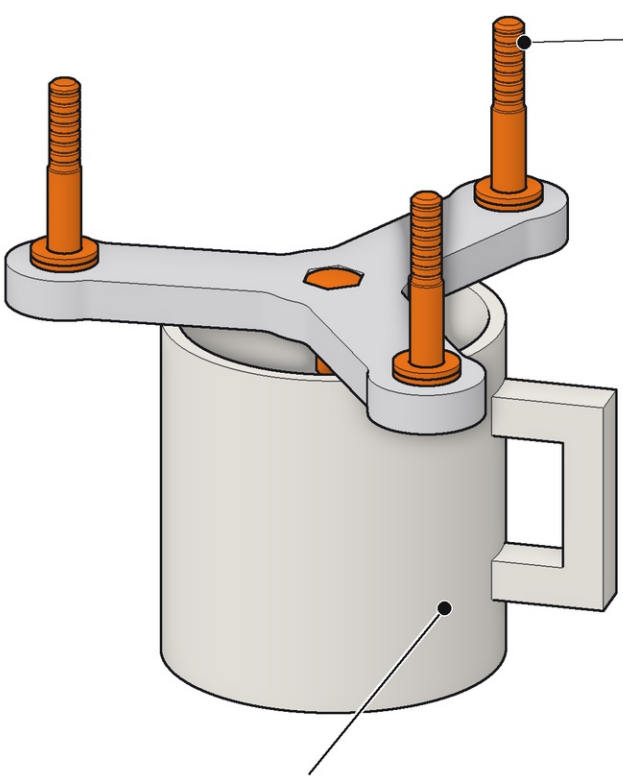


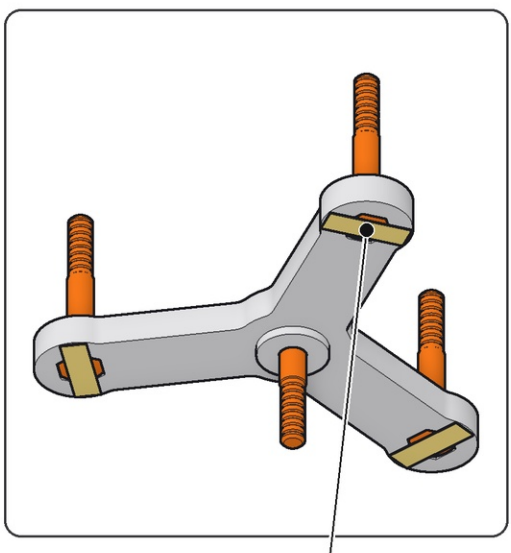
4 mm hex key. Drive slowly: the bolts cut their own thread. Stop when the flange is tight. The wires can point any way round; up is easiest.

Step 4

Bolts into the spider, then two washers on each pin.







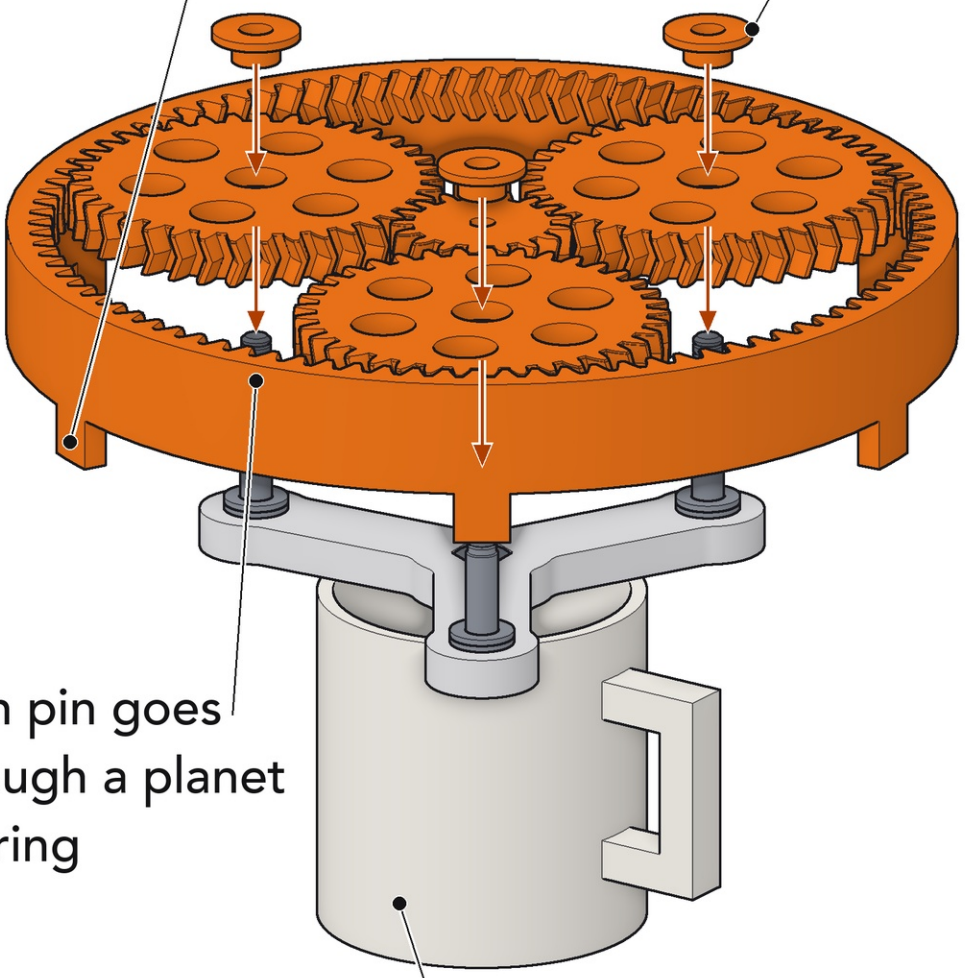
underneath: masking tape over each bolt head stops the bolts dropping out. Peel it off after step 6.

Step 5

Lower the gearset onto the pins, then the pin spacers.

1. gearset: FRONT face down, dogs point DOWN

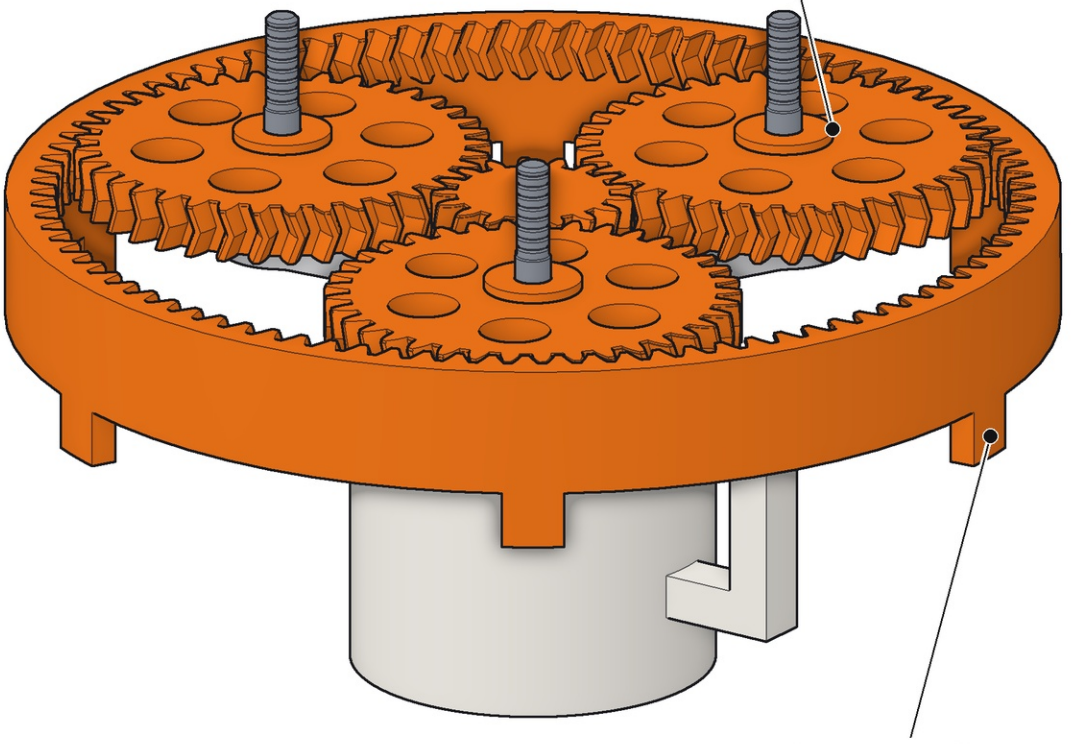
2. pin spacer, x3: wide foot on top



each pin goes through a planet bearing

spider still on the mug

wide foot of each spacer on top



the six dogs point down

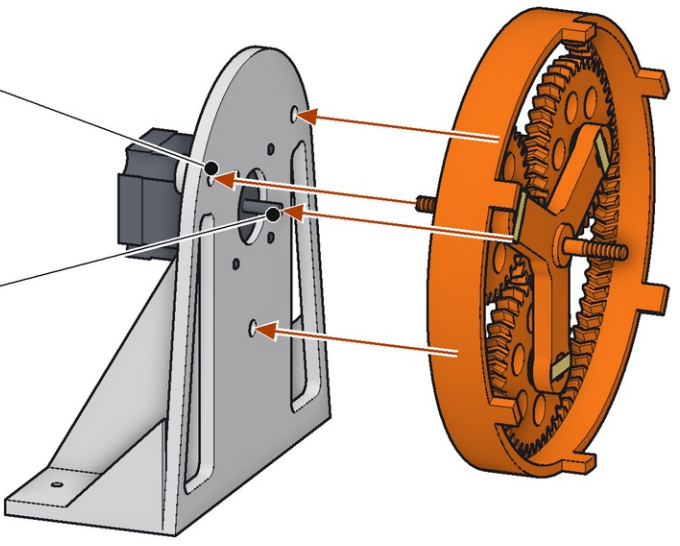
Step 6a

Cartridge onto the wall: pins in, sun onto the shaft.

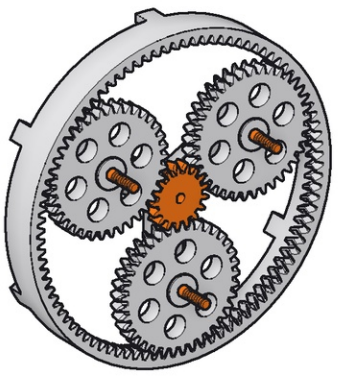
stand the cartridge up, pins toward the wall. Thumbs on the taped bolt heads.

1. three pins into three holes

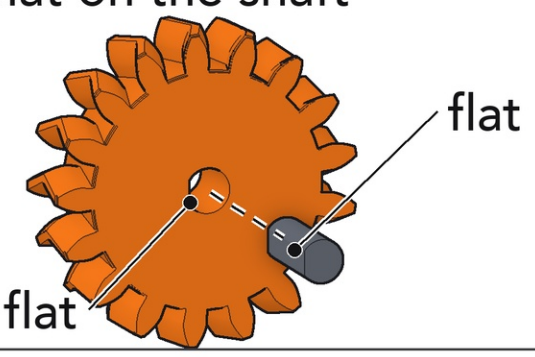
2. sun onto the motor shaft



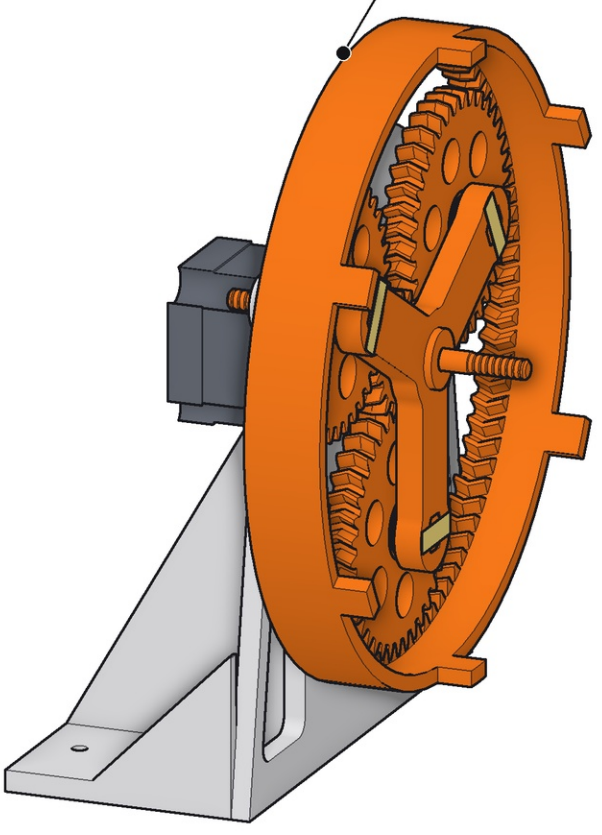
back of the cartridge: 3 pins and the sun



turn the ring until the flat in the sun meets the flat on the shaft



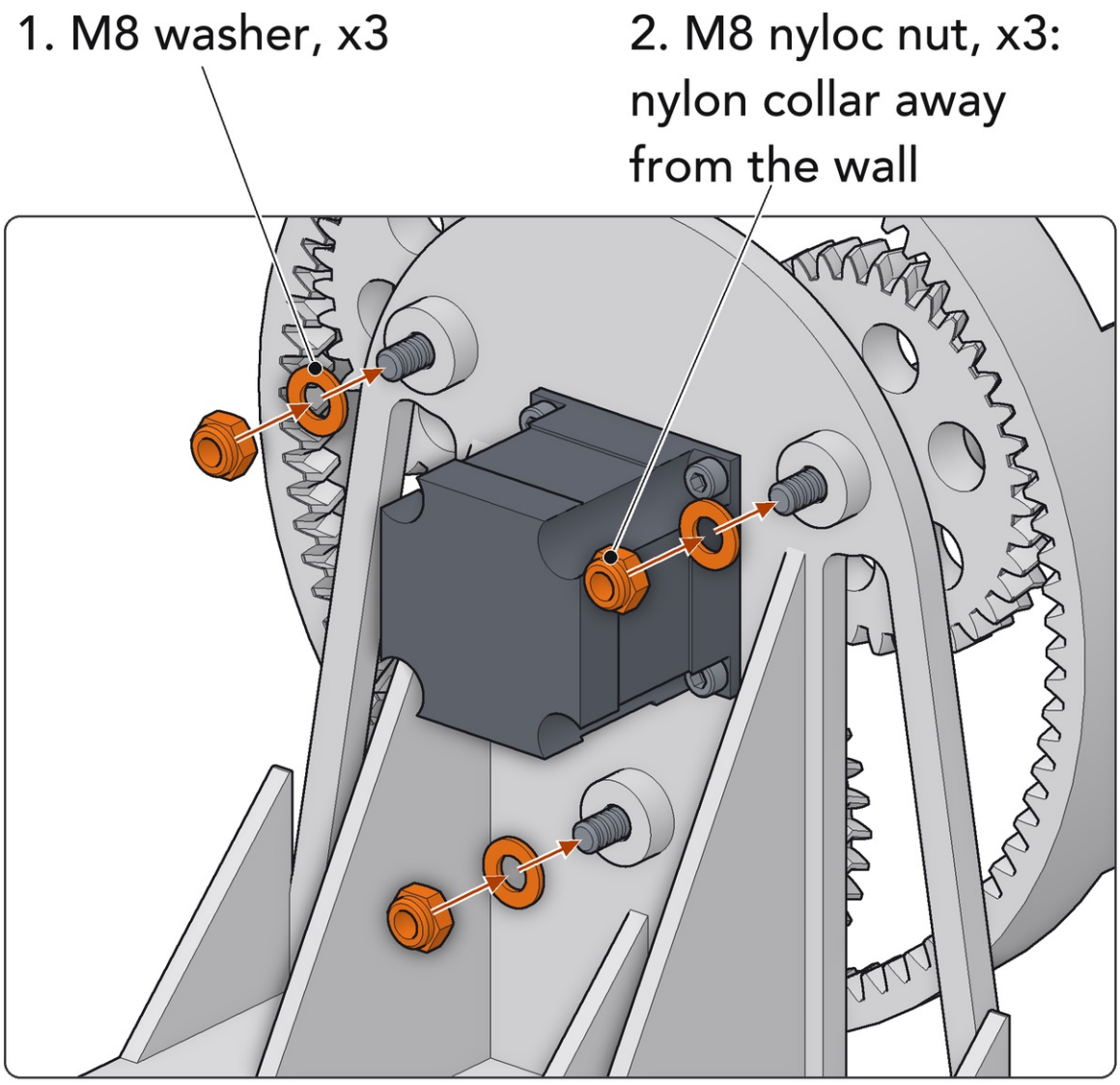
pushed home: only a 3 mm gap between the ring and the wall



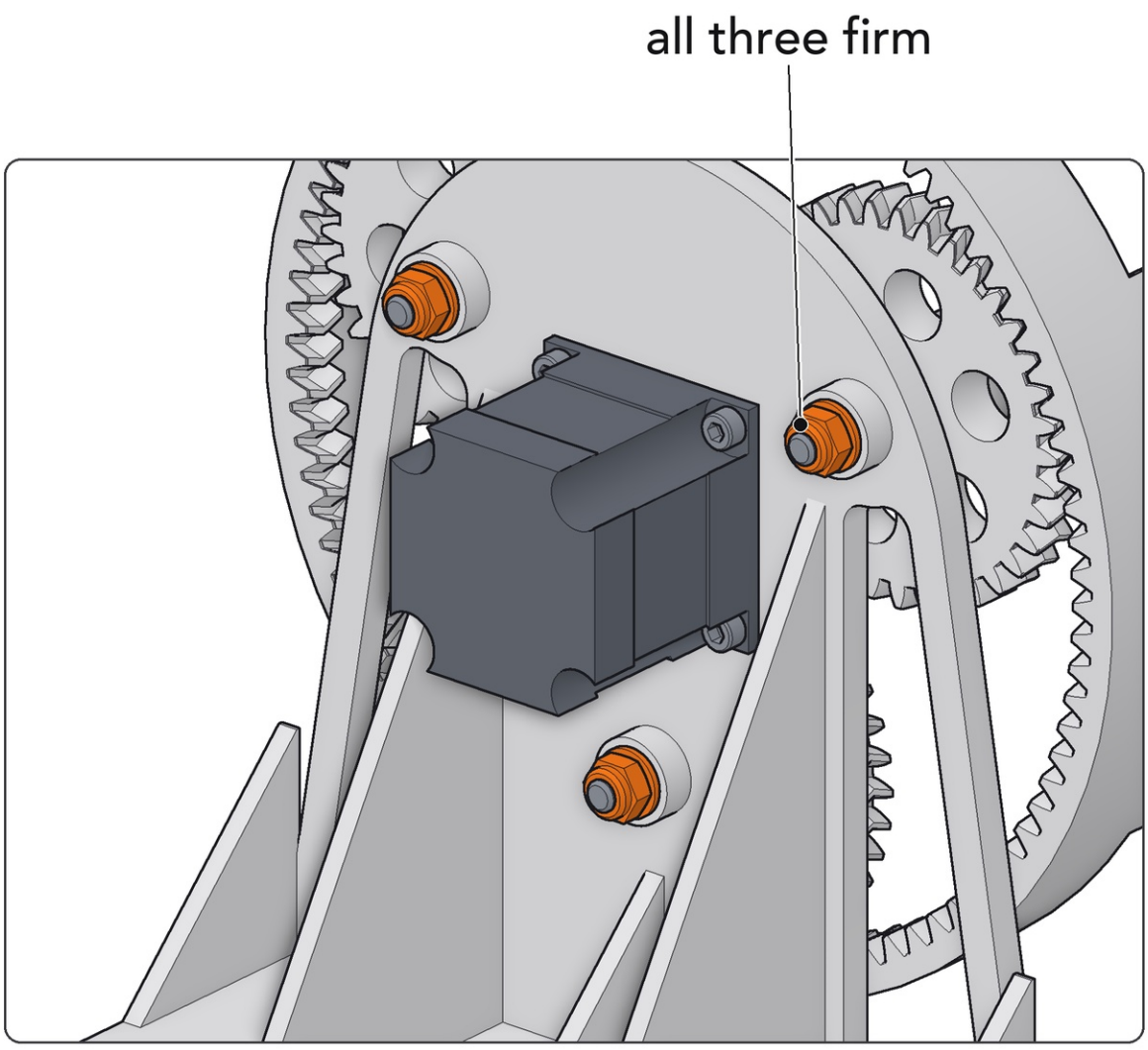
keep a hand on it: nothing holds it until the nuts are on (next picture)

Step 6b

Behind the wall: washer and nyloc nut on each pin.



13 mm spanner. The hex pockets in the spider hold the bolt heads for you.

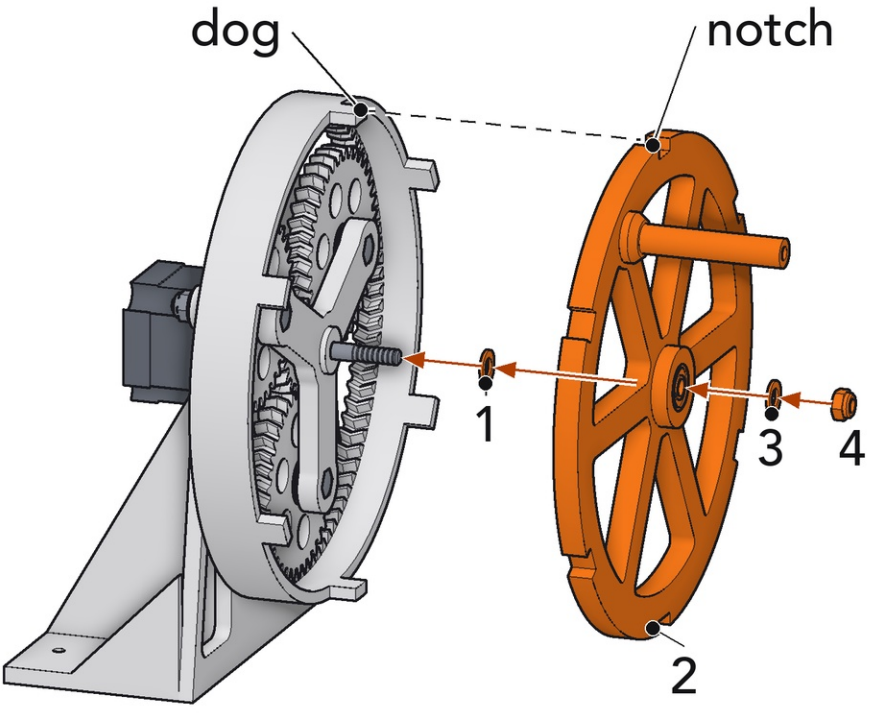


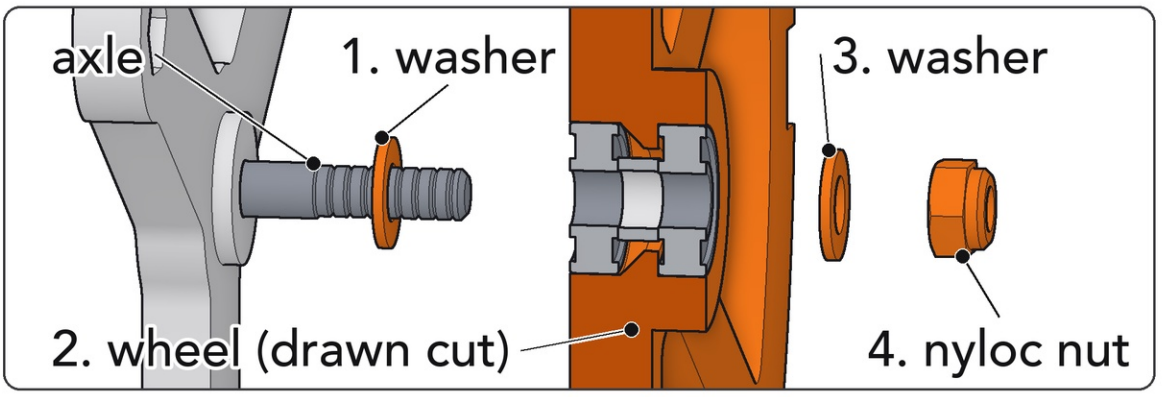
now peel the tape off the three bolt heads at the front

Step 7

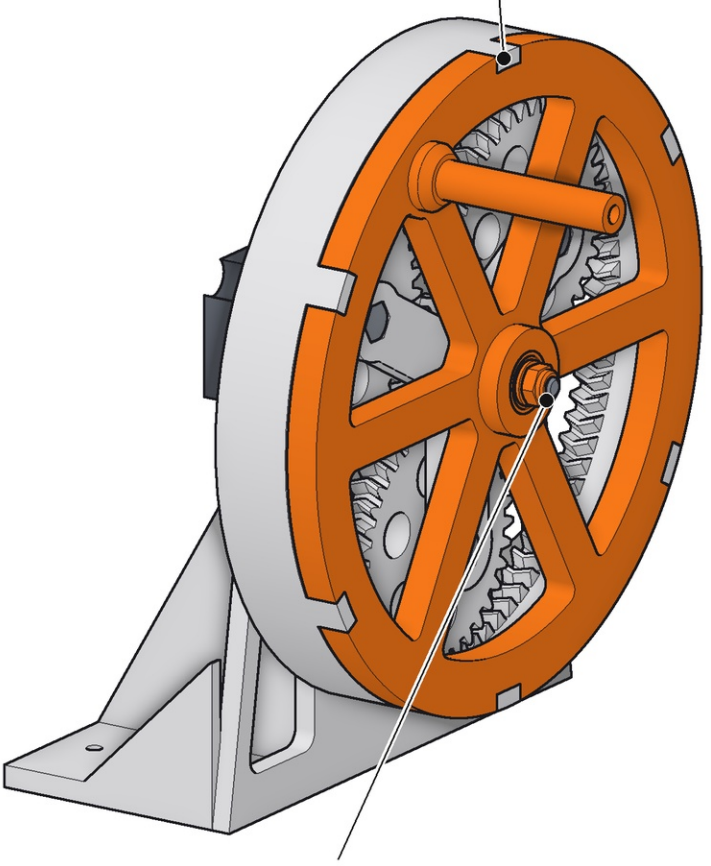
Washer, wheel, washer, nyloc nut onto the axle. Snug only.

turn the wheel until the 6 dogs drop into the 6 notches





all 6 dogs sit in their notches

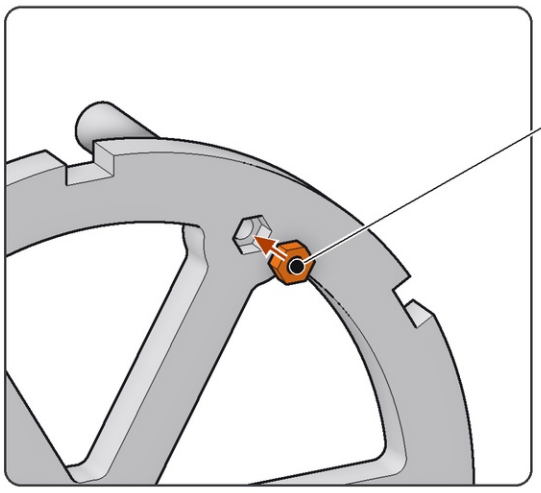


13 mm spanner. Snug, not tight: the wheel must spin freely and the ring should rattle a little.

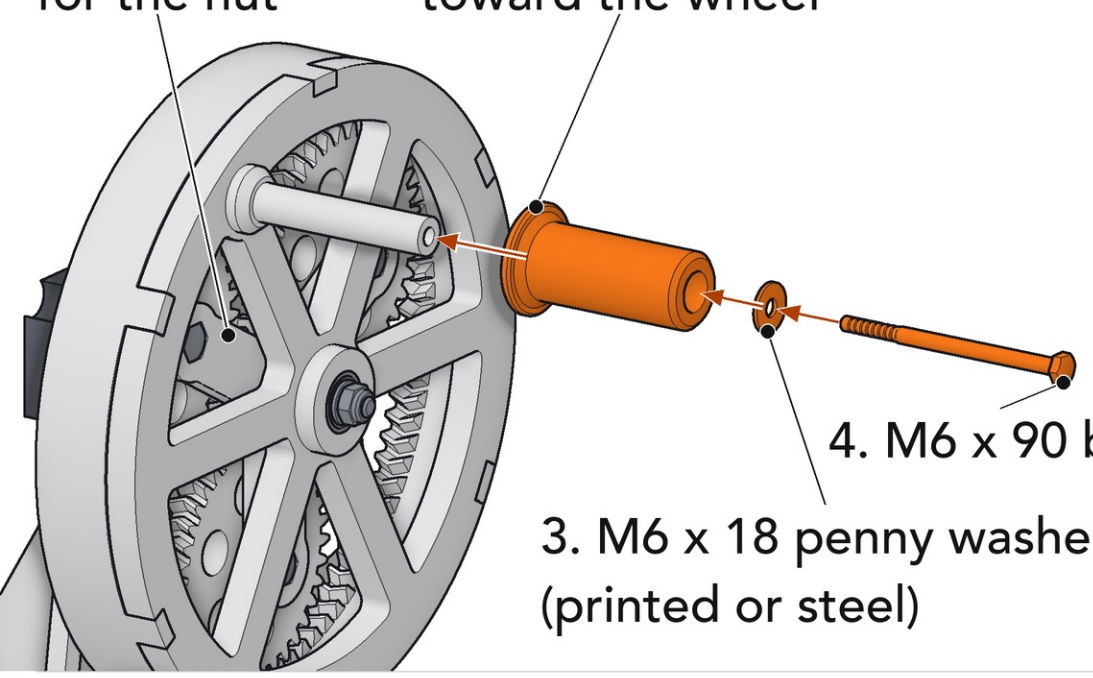
M8 washers used: 6 + 3 + 2 = 11. One is spare.

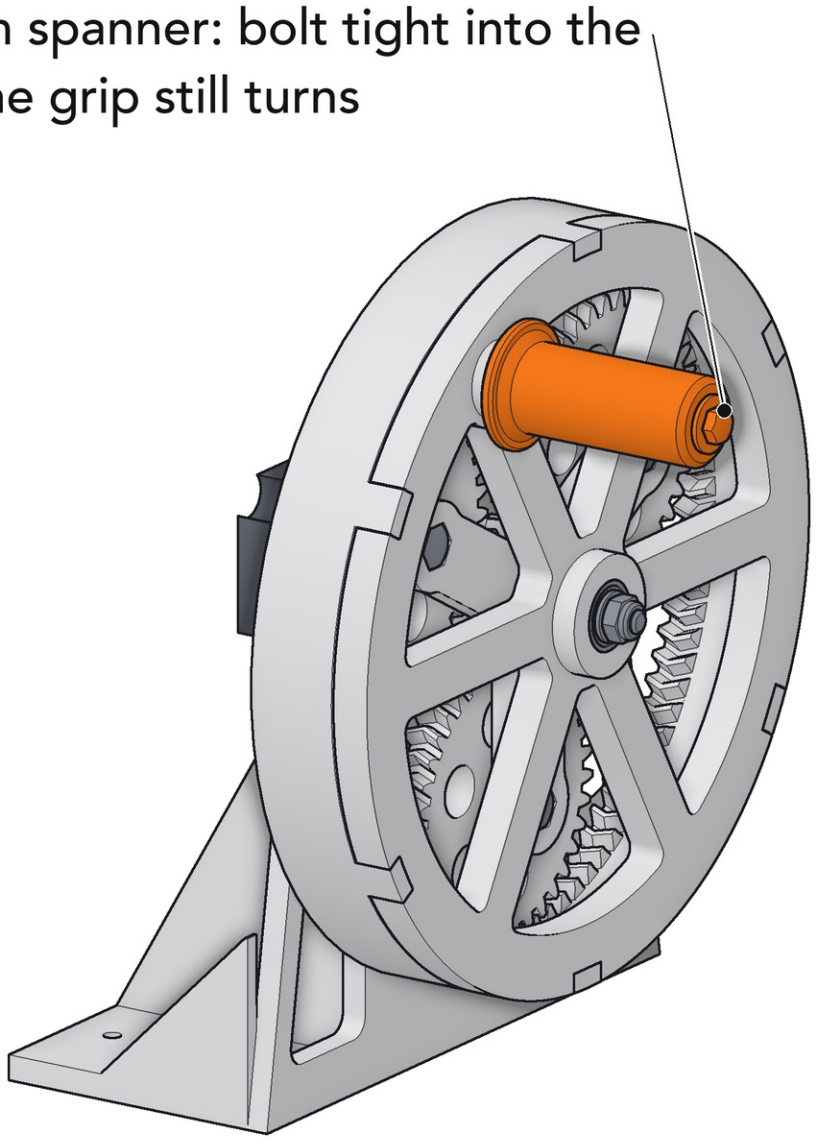
Step 8

Handle: M6 nut behind, then grip, penny washer, bolt.



1. M6 nut into the hex pocket behind the post. Wheel drawn alone, seen from the wall side: yours stays on the machine. Turn the post to the top; fingertip on the nut.



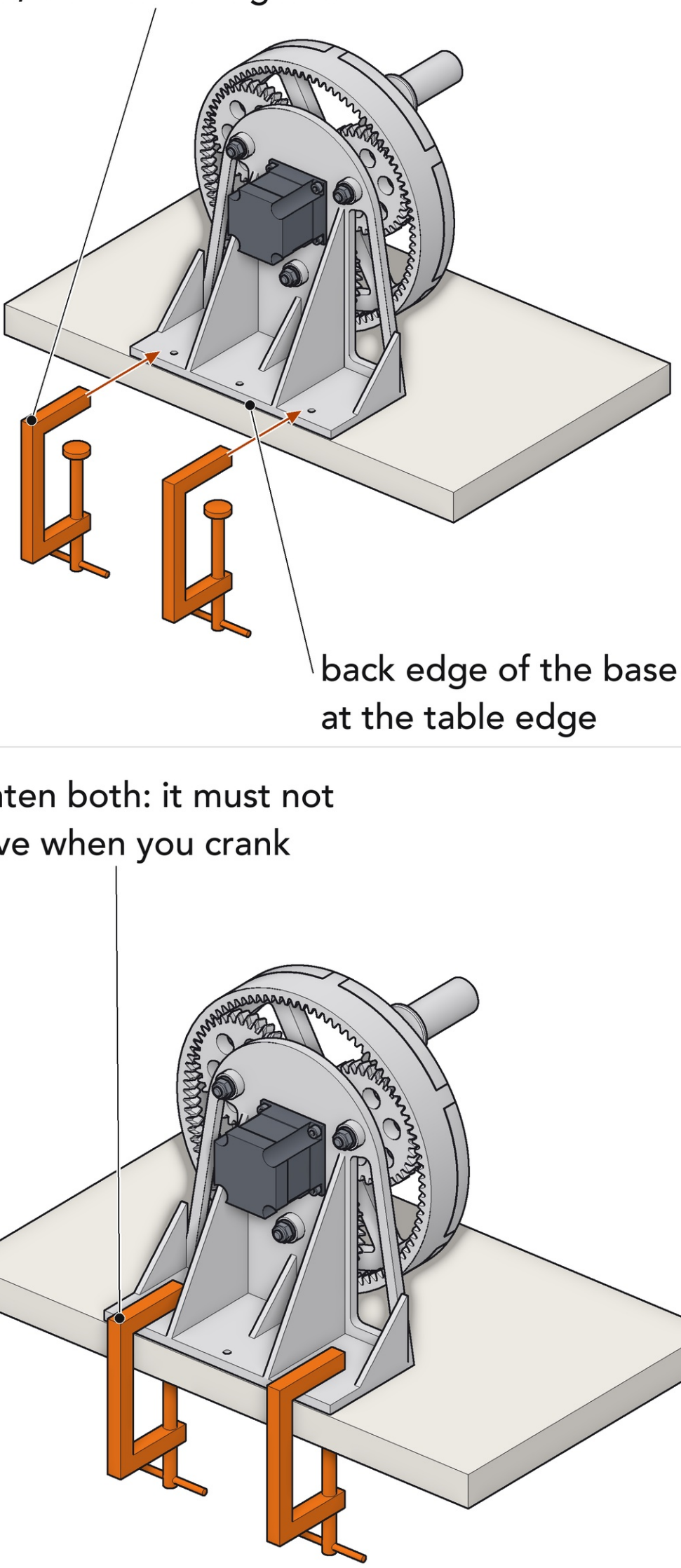


Step 9

Clamp the base to the table with two G-clamps.

seen from behind (the motor side)

G-clamp (100 mm), x2: jaw on the base, between the gussets



back edge of the base at the table edge

tighten both: it must not move when you crank

use a corner of the table, so you can stand beside the wheel to crank